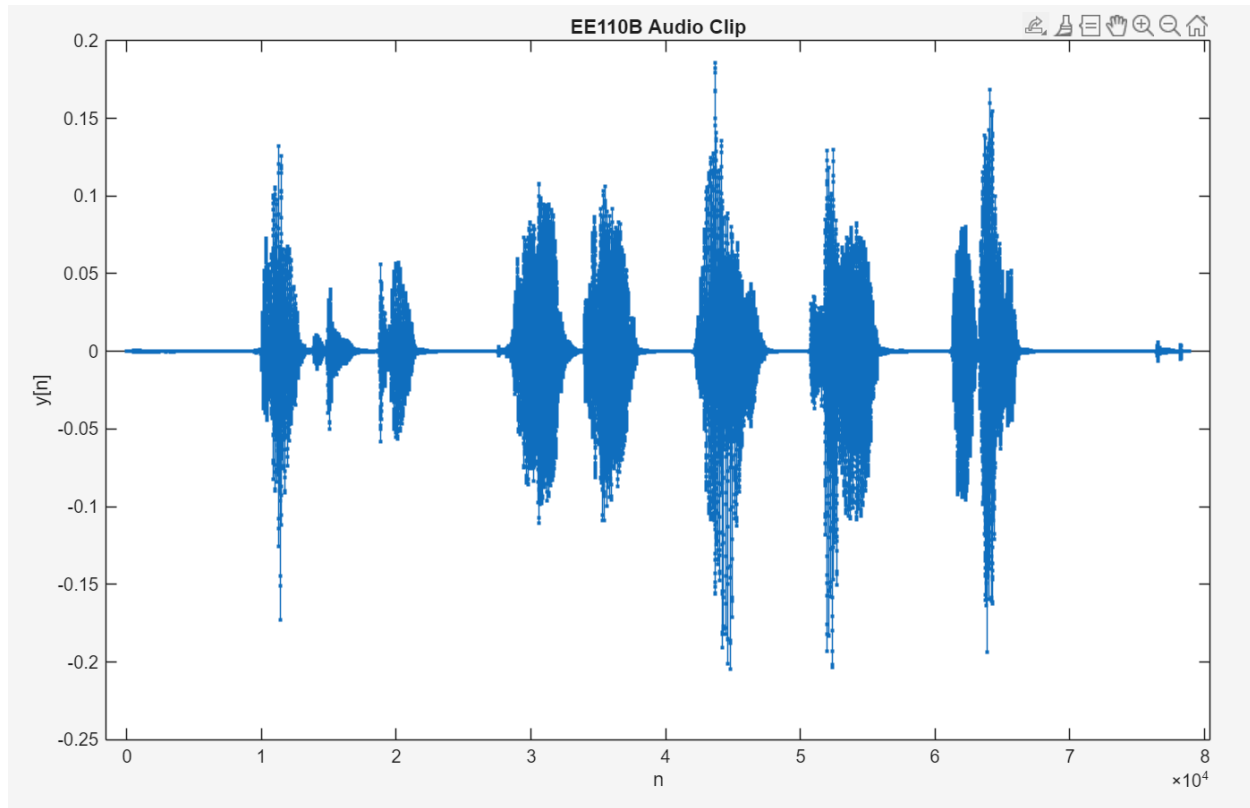


EE110B Lab#2

Nathan Taing

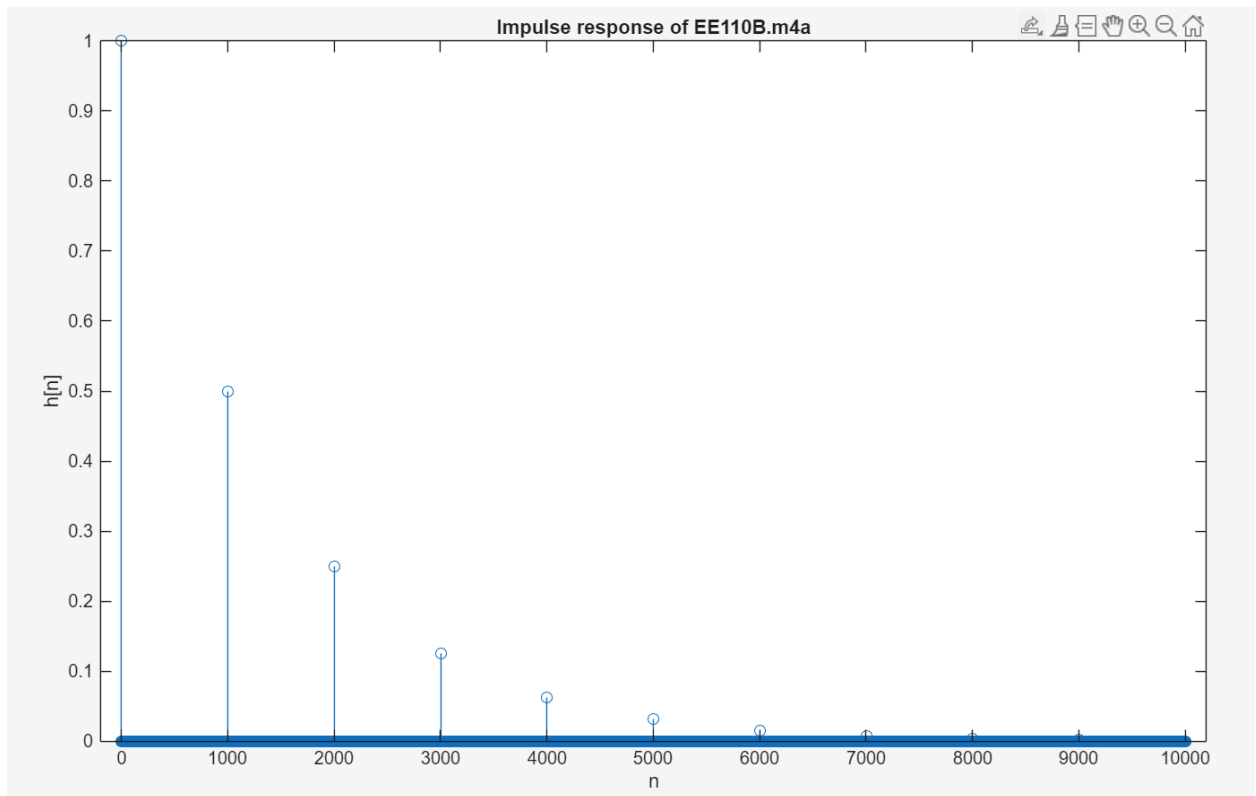
Simulating Reverberations

a)



- b) The impulse response is $h[n] = \sum_{k=0}^{N_{reverb}-1} a_{q_k} \delta[n - q_k]$. We simply use $y[n] = h[n]$ and $x[n] = \delta[n]$ to find the impulse response. The system is stable because there is a finite number of nonzero terms. We can see from the summation that it does not blow up infinitely. Therefore, we know this system is BIBO stable.

c)



Code:

```
h_reverb = zeros(10000, 1);
```

```
for k = 0:9
```

```
    qk = 1000*k;
```

```
    aqk = 1/(2^k);
```

```
    h_reverb(qk + 1) = aqk;
```

```
end
```

```
figure
```

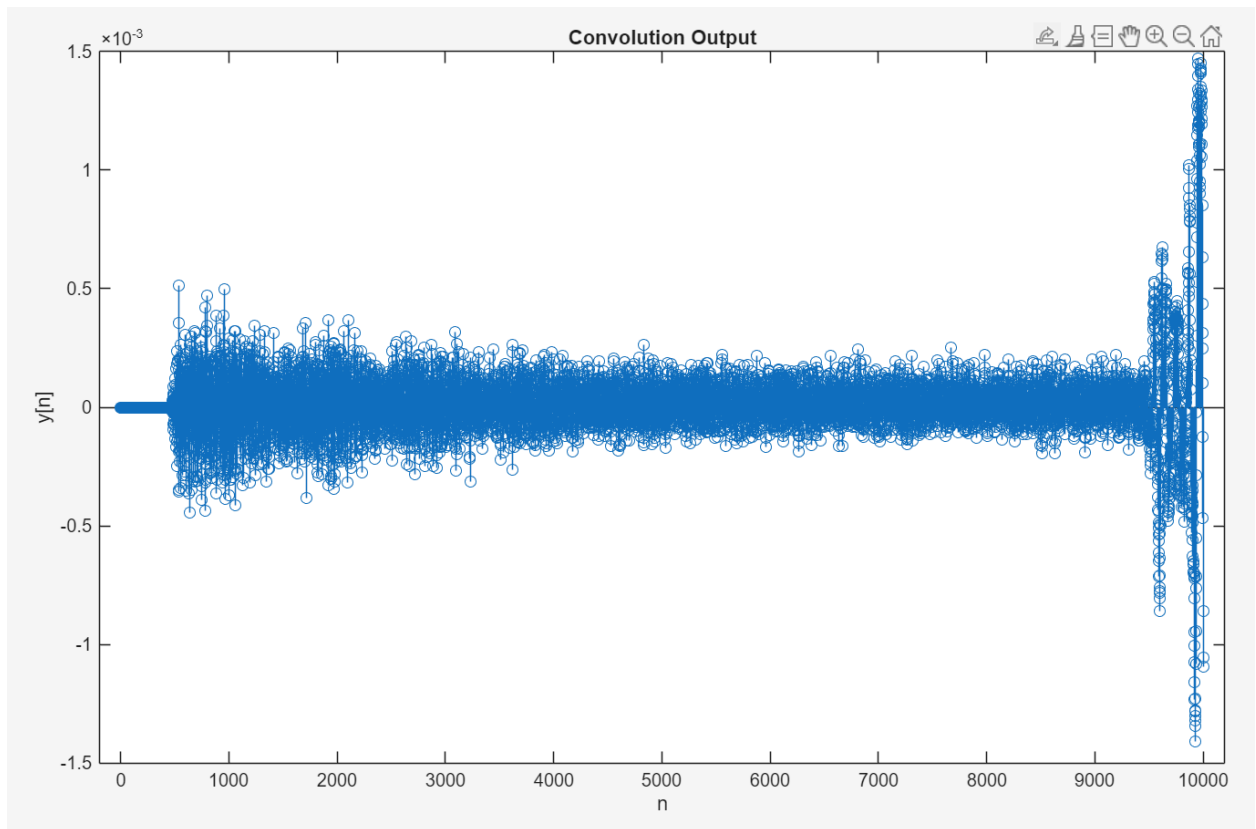
```
stem(0:9999, h_reverb);
```

```
xlabel('n');
```

```
ylabel('h[n]')
```

```
title('Impulse response of EE110B.mp4')
```

d)

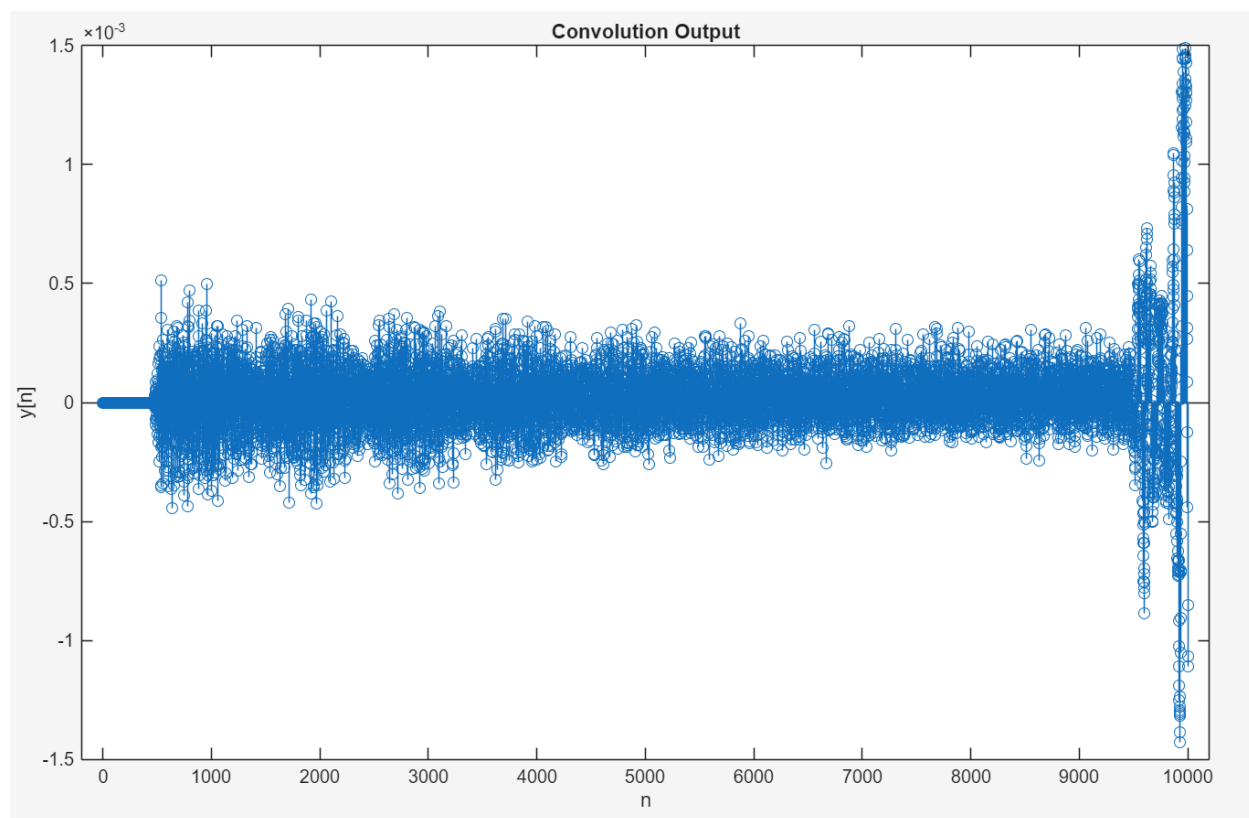
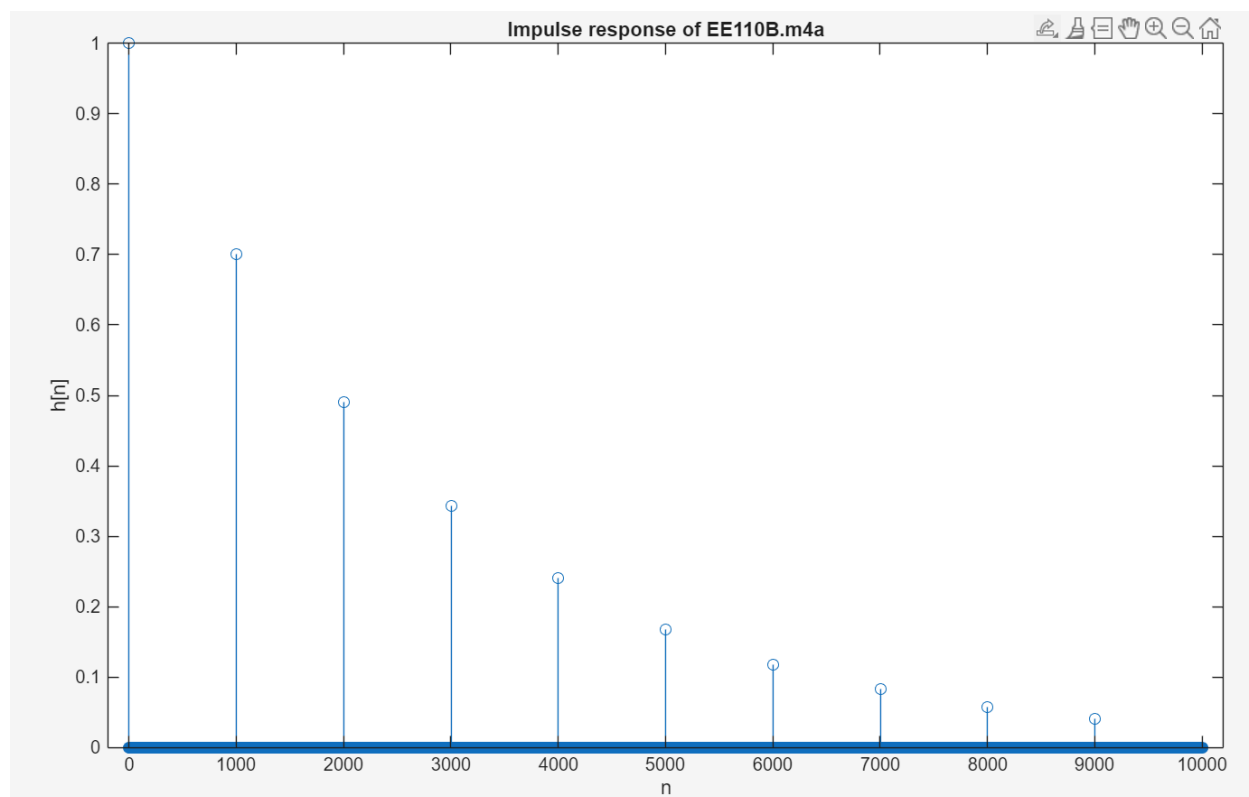


Code:

```
n = 1:10000;
h_reverb = zeros(10000, 1);
for k = 0:9
    qk = 1000*k;
    aqk = 1/(2^k);
    h_reverb(qk + 1) = aqk;
end

figure
reverb_audio = conv(audio_clip, h_reverb);
reverb_audio_y = reverb_audio(1:10000);
stem(n, reverb_audio_y);
xlabel('n');
ylabel('y[n]');
title('Convolution Output');
```

e)

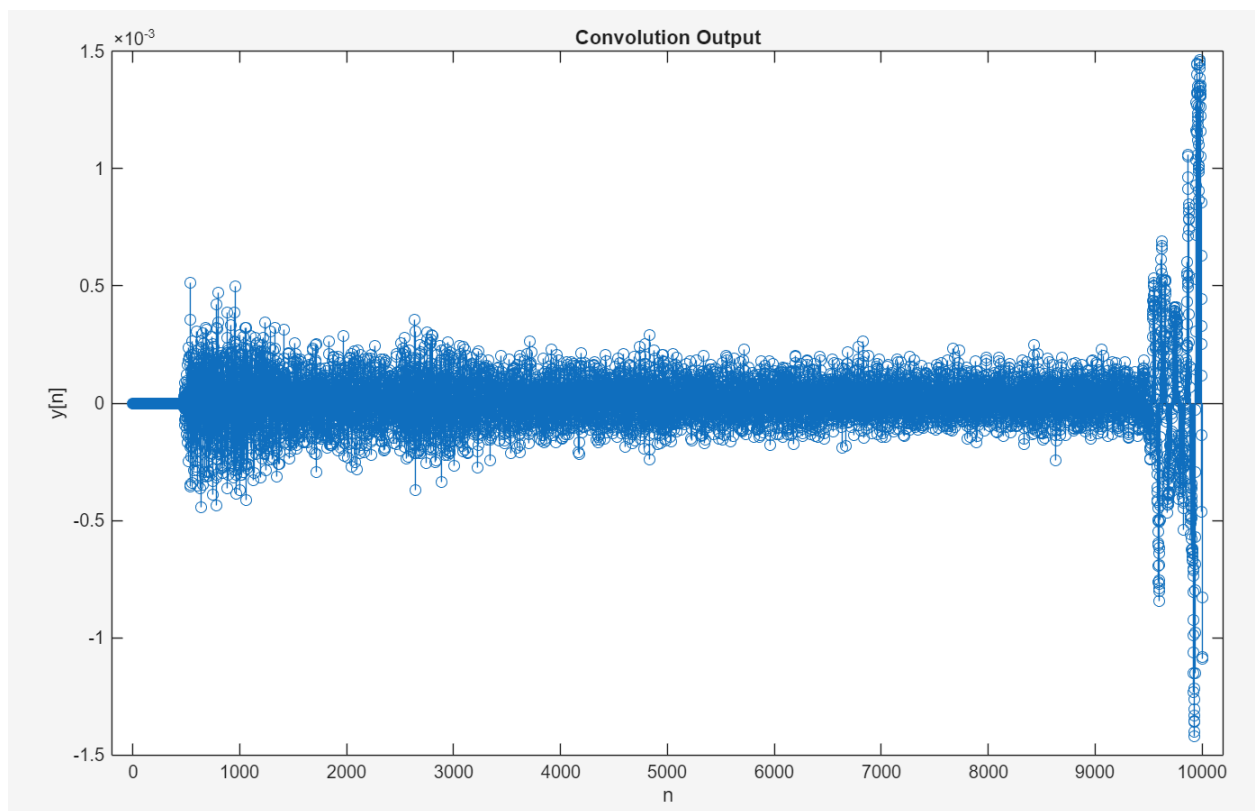
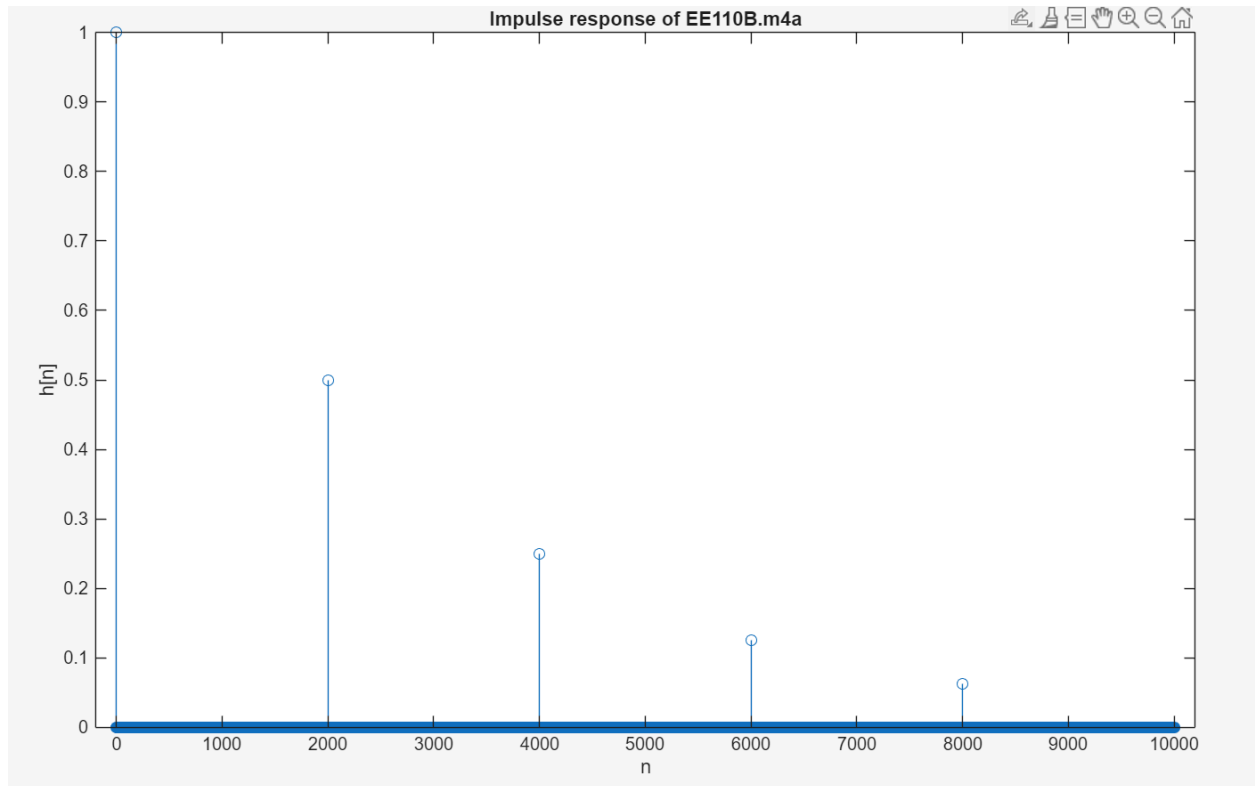


Code:

```
n = 1:10000;
h_reverb = zeros(10000, 1);
for k = 0:9
    qk = 1000*k;
    aqk = 0.7^k;
    h_reverb(qk + 1) = aqk;
end
figure
stem(1:10000, h_reverb);
xlabel('n');
ylabel('h[n]')
title('Impulse response of EE110B.mp4')
figure
reverb_audio = conv(audio_clip, h_reverb);
reverb_audio_y = reverb_audio(1:10000);
stem(n, reverb_audio_y);
xlabel('n');
ylabel('y[n]');
title('Convolution Output');
```

For the impulse response, the difference is a little more obvious as the values of each response is greater than the impulse response of $a_{q_k} = 1/2^k$. However, the convolution outputs look roughly the same. The most glaring differences are that the general shape as the output is a little wider. Otherwise, the shape of both convolution outputs are generally the same. The specific values however, are slightly different.

f)



Code:

```
n = 1:10000;
```

```

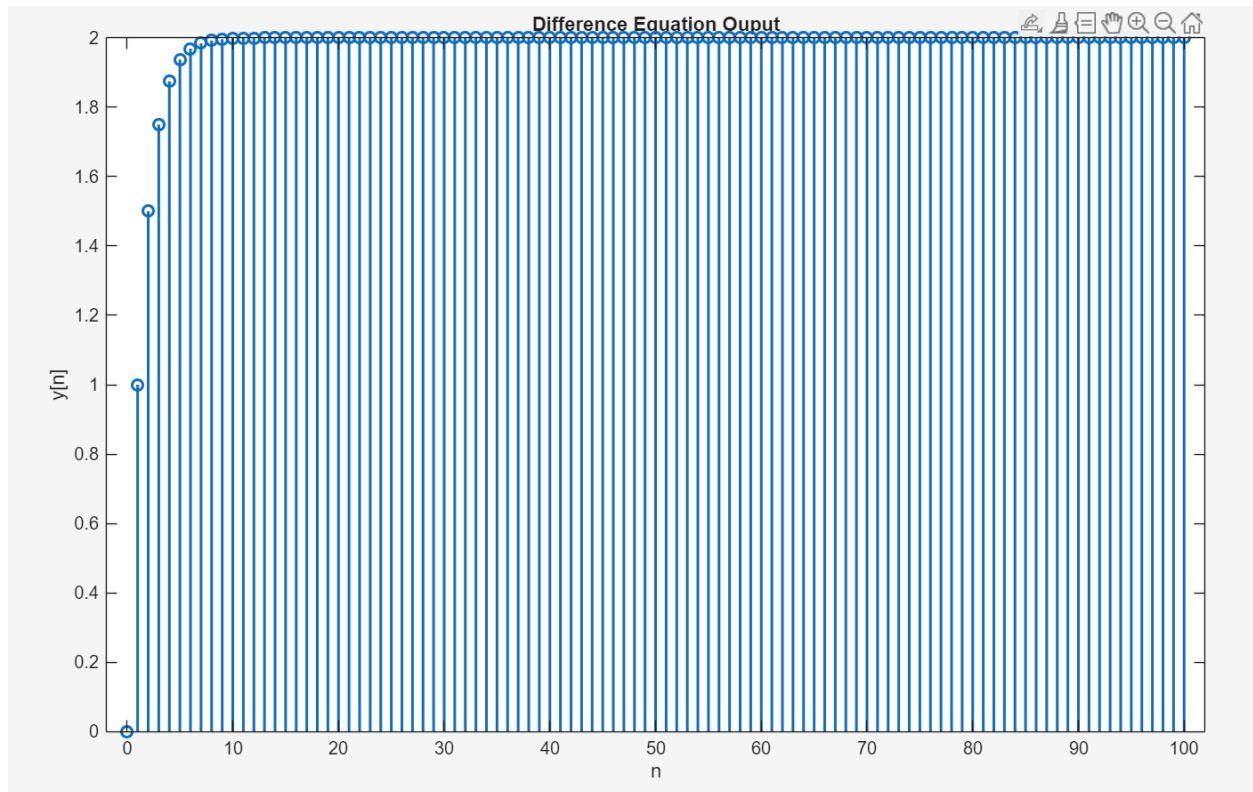
h_reverb = zeros(10000, 1);
for k = 0:5
    qk = 2000*k;
    aqk = (1/2^k);
    h_reverb(qk + 1) = aqk;
end
figure
h_reverb = h_reverb(1:10000);
stem(1:10000, h_reverb);
xlabel('n');
ylabel('h[n]')
title('Impulse response of EE110B.m4a')
figure
reverb_audio = conv(audio_clip, h_reverb);
reverb_audio_y = reverb_audio(1:10000);
stem(n, reverb_audio_y);
xlabel('n');
ylabel('y[n]');
title('Convolution Output');

```

The impulse response is recorded less frequently as there are now five points instead of ten, which is the amount the original contained. The values, however, equal part c. This makes sense as the calculation values are the same, but the frequency we are documenting the samples are different (qk). The convolution also looks to have the same shape, but slightly different cluster of signals.

Difference Equations

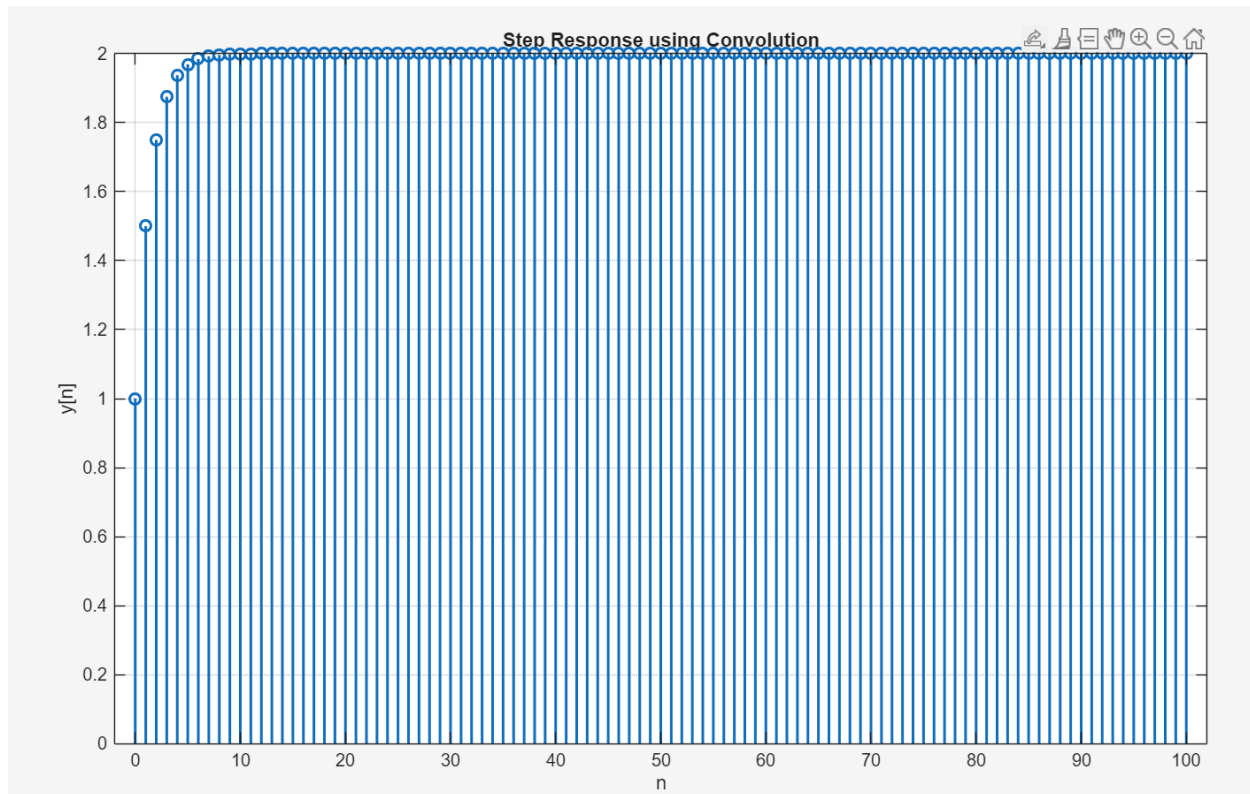
a)



Code:

```
n = 0:100;      % Time axis
x = ones(1, 100); % Input signal
y = zeros(1, 100); % Output
for idx = 1:length(n)
    if idx == 1
        y(1) = 0;
    else
        y(idx) = 1/2 * y(idx-1) + 1;
    end
end
stem(n, y, 'lineWidth', 1.5)
xlabel('n');
ylabel('y[n]');
title('Difference Equation Output');
```

b)



```

n = 0:100;           % time axis
u = ones(1, 101);    % step input u[n]
h = (0.5).^n;         % impulse response h[n]
x = conv(u, h);       % convolution
y = x(1:length(n));
figure
stem(n, y, 'LineWidth', 1.5)
xlabel('n')
ylabel('y[n]')
title('Step Response using Convolution')
grid on

```

- c) Yes, the systems in (a) and (b) are the same. We can obtain $h[n]$ from the differential equation (a) by utilizing $y[n] = h[n]$ and $x[n] = \delta[n]$. Then, we can use multiple n s (ie $n=1, n=2, n=3$, etc.) to find the pattern that leads to $h[n]$. After solving for multiple n s, we will see the pattern of $(\frac{1}{2})^n$, which is what $h[n]$ equals.